Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 07: Python Programming III

Digital for Executive

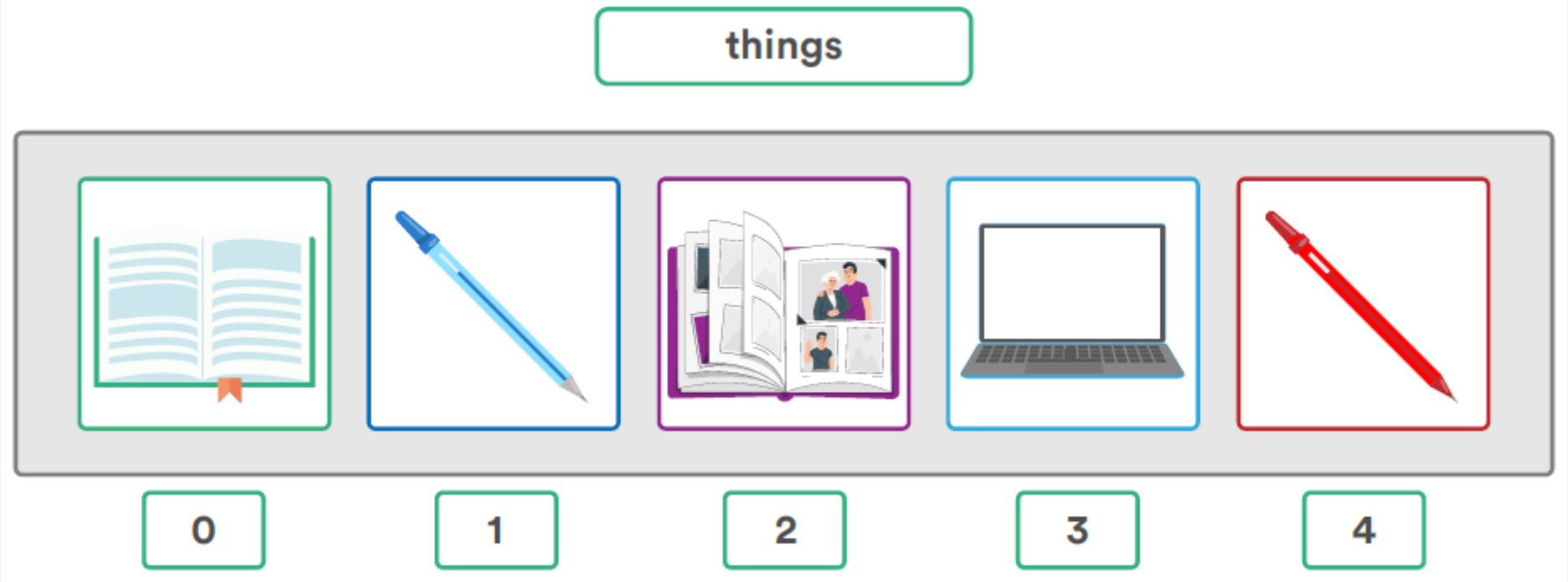
Outline

- Python Collection Data Types (Part 2)
- Python Function
- Exercise

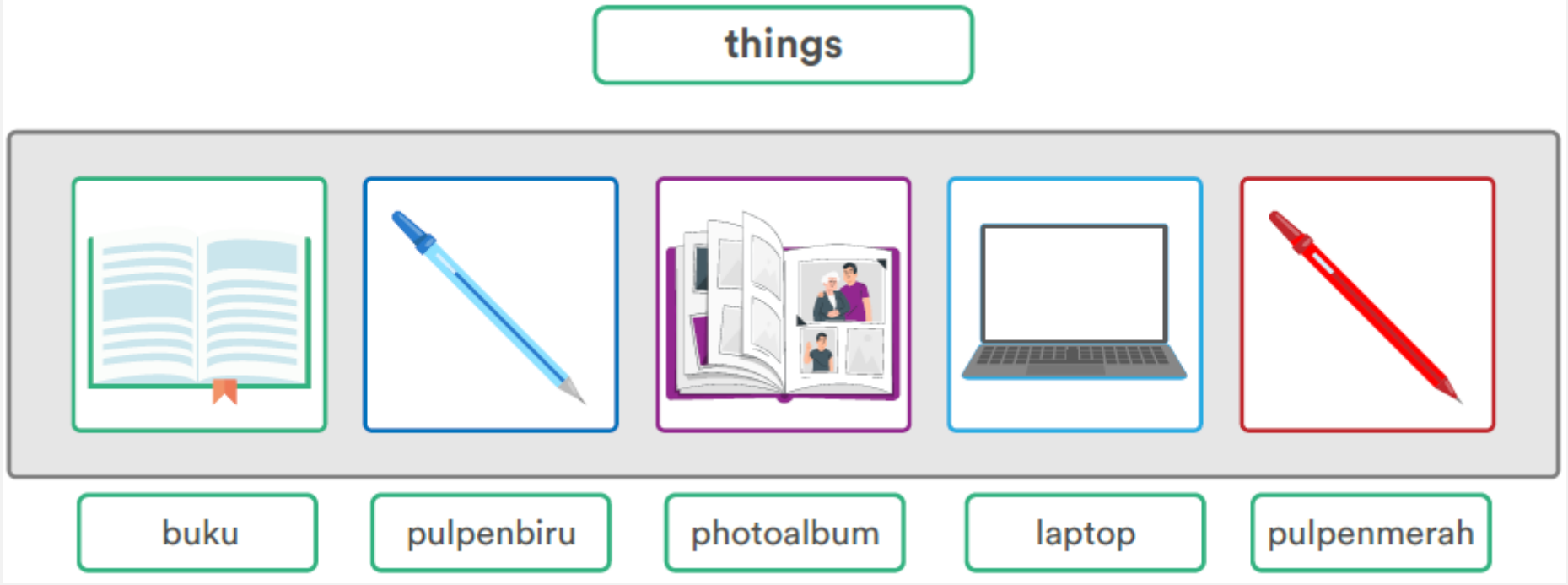
Python Collection Data Types (Part 2)

Dictionary

Dictionary Introduction



Dictionary Introduction

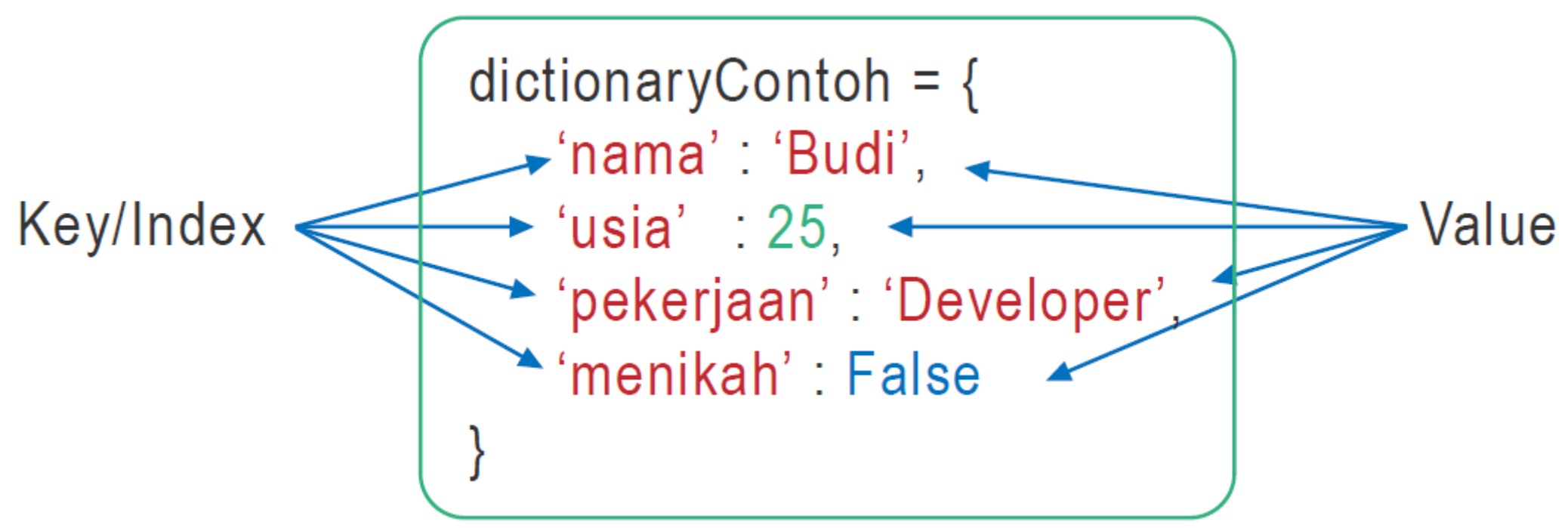


Dictionary Introduction

- What is Dictionary?
- Why?
- When?

Syntax Dictionary

Syntax Dictionary



Create Dictionary

```
# cara 1
dictionaryContoh1  = {
      'nama' : 'Budi',
      'usia'     : 25,
      'pekerjaan' : 'Developer'
}
print(dictionaryContoh1)   #  {'nama': 'Budi', 'usia': 25, 'pekerjaan': 'Developer'}

# cara 2
dictionaryContoh2  = dict(nama='Andi', usia=27, pekerjaan='Data Scientist')
print(dictionaryContoh2)   #  {'nama': 'Andi', 'usia': 27, 'pekerjaan': 'Data Scientist'}
```

Access Data

```
dictionaryContoh = {
    'nama' : 'Budi',
    'usia'     : 25,
    'pekerjaan' : 'Developer',
    'menikah' : False
}

print(dictionaryContoh['nama'])  #  Budi
print(dictionaryContoh['usia'])  #  25
print(dictionaryContoh.get('pekerjaan'))  #  Developer
print(dictionaryContoh.get('menikah'))  #  False
```

Change Data

```
dictionaryContoh = {
    'nama' : 'Budi',
    'usia'    : 25,
    'pekerjaan' : 'Developer',
    'menikah' : False
}

dictionaryContoh['usia']  =  27
print(dictionaryContoh)

#  {'nama': 'Budi', 'usia': 27, 'pekerjaan': 'Developer', 'menikah': False}
```

Add Data

```
dictionaryContoh = {
    'nama' : 'Budi',
    'usia'    : 25,
    'pekerjaan' : 'Developer',
    'menikah' : False
}

dictionaryContoh['kelamin']  =  'Pria'
print(dictionaryContoh)

#  {'nama': 'Budi', 'usia': 25, 'pekerjaan': 'Developer', 'menikah': False, 'kelamin': 'Pria'}
```

Delete Data

```
dictionaryContoh = {
        'nama' : 'Budi',
        'usia' : 25,
        'pekerjaan' : 'Developer',
        'menikah' : False,
        'kelamin'  : 'Pria'
}

del  dictionaryContoh['kelamin']  # removes key-value pair for kelamin key
dictionaryContoh.pop('usia')  # removes key-value pair for usia key
dictionaryContoh.popitem( )  # removes the last inserted item

print(dictionaryContoh)  #  {'nama': 'Budi', 'pekerjaan': 'Developer'}

dictionaryContoh.clear( )  # empties the dictionary

print(dictionaryContoh)  #  { }
```

Loop Through a Dictionary

```
dictionaryContoh = {
        'nama' : 'Budi',
        'usia'       : 25,
        'pekerjaan' : 'Developer',
        'menikah' : False
}

for key in dictionaryContoh :
        print("{} = {}".format(key,dictionaryContoh[key]))

for key in dictionaryContoh.keys() :
        print("{} = {}".format(key,dictionaryContoh[key]))

for val in dictionaryContoh.values() :
        print(val)

for key,val in dictionaryContoh.items() :
        print("{} = {}".format(key,val))
```

Check if a Key Exists in a Dictionary

```
dictionaryContoh = {
        'nama' : 'Budi',
        'usia'       : 25,
        'pekerjaan' : 'Developer',
        'menikah' : False
}

if 'usia' in dictionaryContoh :
        print('Ada key usia pada dictionaryContoh')

else :
        print('Tidak ada key usia pada dictionaryContoh')

# Ada key usia pada dictionaryContoh
```

Length of Dictionary

```
dictionaryContoh = {
    'nama' : 'Budi',
    'usia'     : 25,
    'pekerjaan' : 'Developer',
    'menikah' : False
}

print(len(dictionaryContoh))  #  4
```

Copy a Dictionary

```
dictionaryContoh = {
        'nama' : 'Budi',
        'usia'       : 25,
        'pekerjaan' : 'Developer',
        'menikah' : False
}

newDictionary1 = dictionaryContoh
newDictionary1['nama'] = 'Andi'

newDictionary2 = dictionaryContoh.copy()
newDictionary2['pekerjaan'] = 'Guru'

print(newDictionary1)  #  {'nama': 'Andi', 'usia': 25, 'pekerjaan': 'Developer', 'menikah': False}
print(newDictionary2)  #  {'nama': 'Andi', 'usia': 25, 'pekerjaan': 'Guru', 'menikah': False}
print(dictionaryContoh)  #  {'nama': 'Andi', 'usia': 25, 'pekerjaan': 'Developer', 'menikah': False}
```

Dictionary inside Dictionary

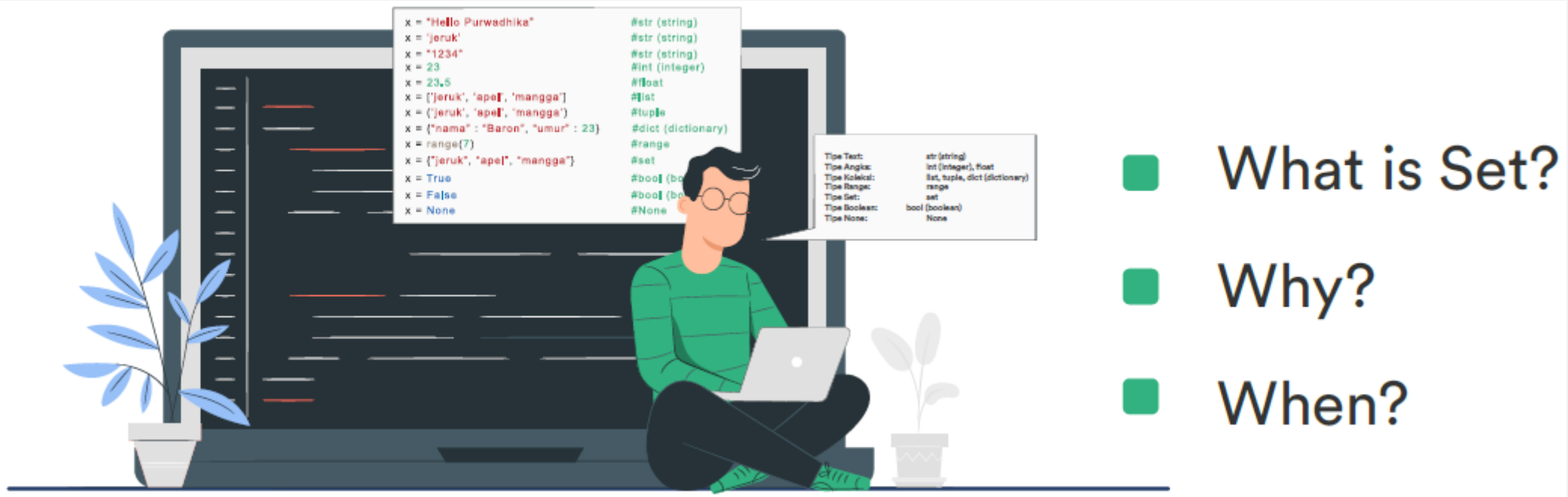
```
things = {
    'food1' : {
        'name'  : 'Ramen',
        'price' : 25000
    },
    'food2' : {
        'name'  : 'Pizza',
        'price' : 30000
    },
    'food3' : {
        'name'  : 'Spaghetti',
        'price' : 20000
    }
}
```

things['food1'] => {'name': 'Ramen', 'price': 25000}

things['food2']['name'] => 'Pizza'

Set

Set Introduction



Syntax Set

Syntax Set

```
setContoh  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Iron Man 3' }
print(setContoh)

# {'Avengers: Age of Ultron', 'Iron Man 3', 'The Avengers'}
# The order of the items will always be random
```

Create Set from List, Tuple, and Dictionary

```
list1 = ['Budi', 2, 2, 'Ceci']
tuple1 = (False, 1, 'Andi', False)
dictionary1 = {
        'nama' : 'Coder',
        'usia'  : 25,
        'pekerjaan' : 'Coder',
        'menikah' : False
}

setList = set(list1)
setTuple = set(tuple1)
setDictionary = set(dictionary1)
setDictionaryValues = set(dictionary1.values())

print(setList)  #  {'Ceci', 2, 'Budi'}
print(setTuple)  #  {False, 1, 'Andi'}
print(setDictionary)  #  {'nama', 'usia', 'pekerjaan', 'menikah'}
print(setDictionaryValues)  #  {False, 25, 'Coder'}
```

Access Data

```
setContoh  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron' }

for item in setContoh :
      print(item)

# Iron Man 3
# Avengers: Age of Ultron
# The Avengers
```

Check if an Item Exists in a Set

```
setContoh = { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron' }

if 'Iron Man 3' in setContoh :
    print('Ada film Iron Man 3 pada setContoh')
else :
    print('Tidak ada film Iron Man 3 pada setContoh')

# Ada film Iron Man 3 pada setContoh
```

Add Data

```
setContoh  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron' }

setContoh.add('Iron Man')
print(setContoh)
# {'Avengers: Age of Ultron', 'The Avengers', 'Iron Man 3', 'Iron Man'}

setContoh.update({'Iron Man 3', 'Hulk'})
print(setContoh)
# {'Hulk', 'Iron Man 3', 'Avengers: Age of Ultron', 'The Avengers', 'Iron Man'}
```

Delete Data

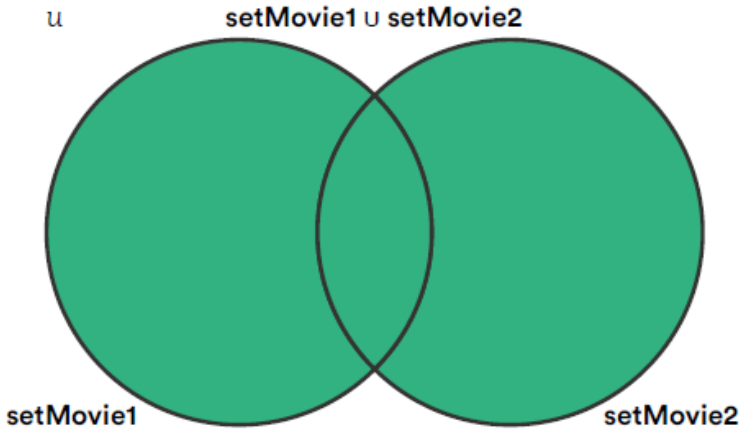
```
setContoh = { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk', 'Wonder Woman' }

setContoh.remove('The Avengers')
print(setContoh)
# {'Wonder Woman', 'Hulk', 'Iron Man 3', 'Avengers: Age of Ultron'}

setContoh.discard('Hulk')
print(setContoh)
# {'Wonder Woman', 'Iron Man 3', 'Avengers: Age of Ultron'}

setContoh.pop()
print(setContoh)
# {'Iron Man 3', 'Avengers: Age of Ultron'}

setContoh.clear()
print(setContoh)
# set()
```

Length of Set

```
setContoh = { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }

print(len(setContoh)) # 4
```

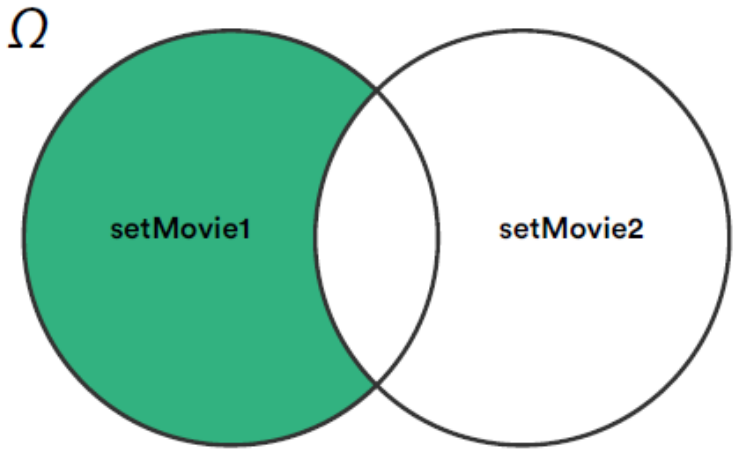
Join Sets

```
setMovie1  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }
setMovie2  =  { 'Titanic', 'The Avengers' }

setGabungan = setMovie1.union(setMovie2)
print(setGabungan)
# {'Iron Man 3', 'Hulk', 'Titanic', 'The Avengers', 'Avengers: Age of Ultron'}
```



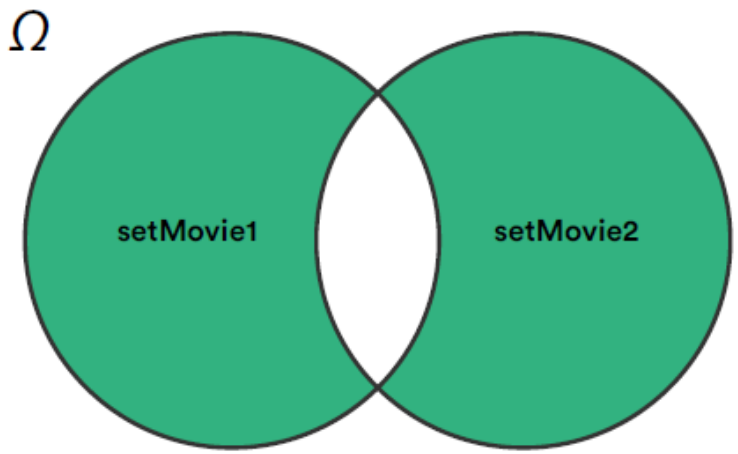
Get difference between 2 sets

```
setMovie1  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }
setMovie2  =  { 'Titanic', 'The Avengers', 'Hulk' }

setMovie3  = setMovie1.difference(setMovie2)
print(setMovie3)
# {'Avengers: Age of Ultron', 'Iron Man 3'}
print(setMovie1)
# {'Iron Man 3', 'Hulk', 'The Avengers', 'Avengers: Age of Ultron'}

setMovie1.difference_update(setMovie2)
print(setMovie1)
# {'Iron Man 3', 'Avengers: Age of Ultron'}
```



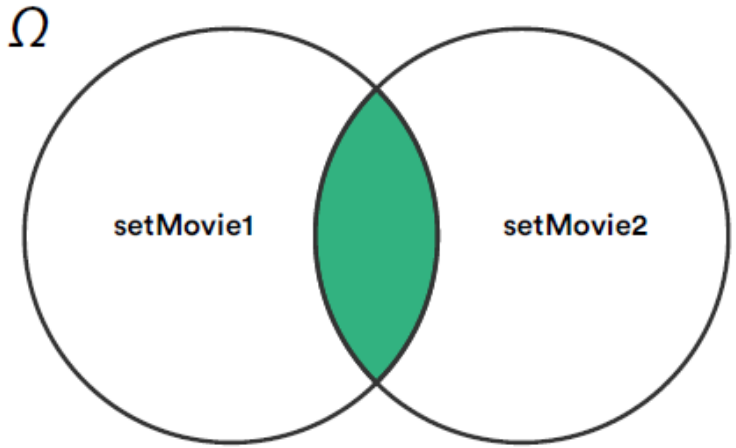
Get symmetric difference between 2 sets

```
setMovie1  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }
setMovie2  =  { 'Titanic', 'The Avengers', 'Hulk' }

setMovie3  = setMovie1.symmetric_difference(setMovie2)
print(setMovie3)
# {'Avengers: Age of Ultron', 'Iron Man 3', 'Titanic'}
print(setMovie1)
# {'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk'}

setMovie1.symmetric_difference_update(setMovie2)
print(setMovie1)
# {'Iron Man 3', 'Avengers: Age of Ultron', 'Titanic'}
```



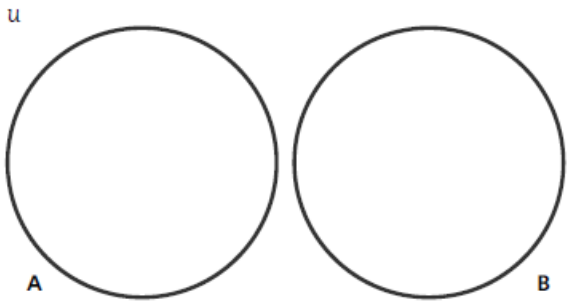
Get intersection between 2 sets

```
setMovie1  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }
setMovie2  =  { 'Titanic', 'The Avengers', 'Hulk' }

setMovie3  = setMovie1.intersection(setMovie2)
print(setMovie3)
# {'The Avengers', 'Hulk'}
print(setMovie1)
# {'Iron Man 3', 'The Avengers', 'Avengers: Age of Ultron', 'Hulk'}

setMovie1.intersection_update(setMovie2)
print(setMovie1)
# {'The Avengers', 'Hulk'}
```

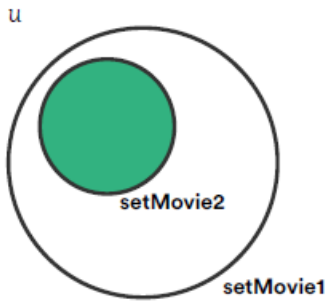


Check disjoint, subset, and superset

```
setMovie1  =  { 'The Avengers', 'Iron Man 3', 'Avengers: Age of Ultron', 'Hulk' }
setMovie2  =  { 'The Avengers', 'Hulk' }

checkDisjoint = setMovie1.isdisjoint(setMovie2)
checkSubset = setMovie2.issubset(setMovie1)
checkSuperset = setMovie1.issuperset(setMovie2)

print(checkDisjoint) # False
print(checkSubset) # True
print(checkSuperset) # True
```



disjoint

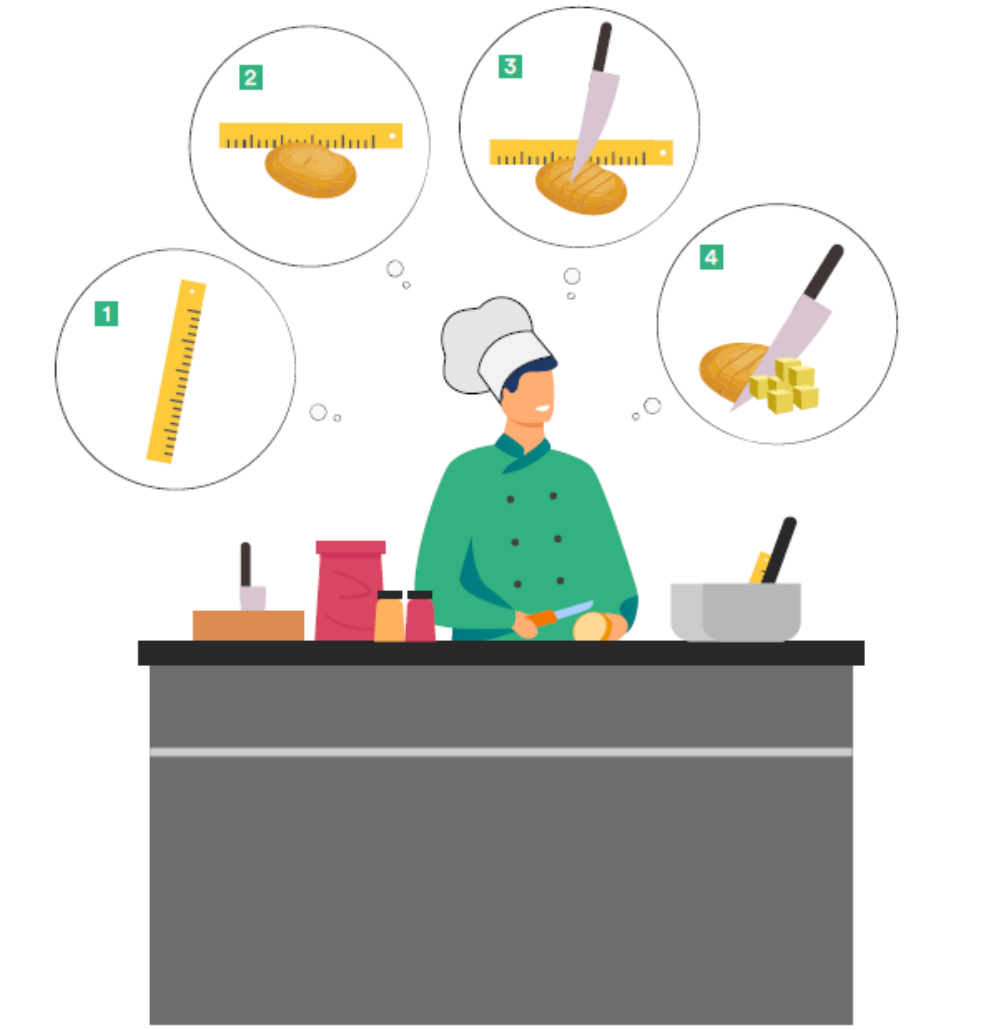


Superset and subset

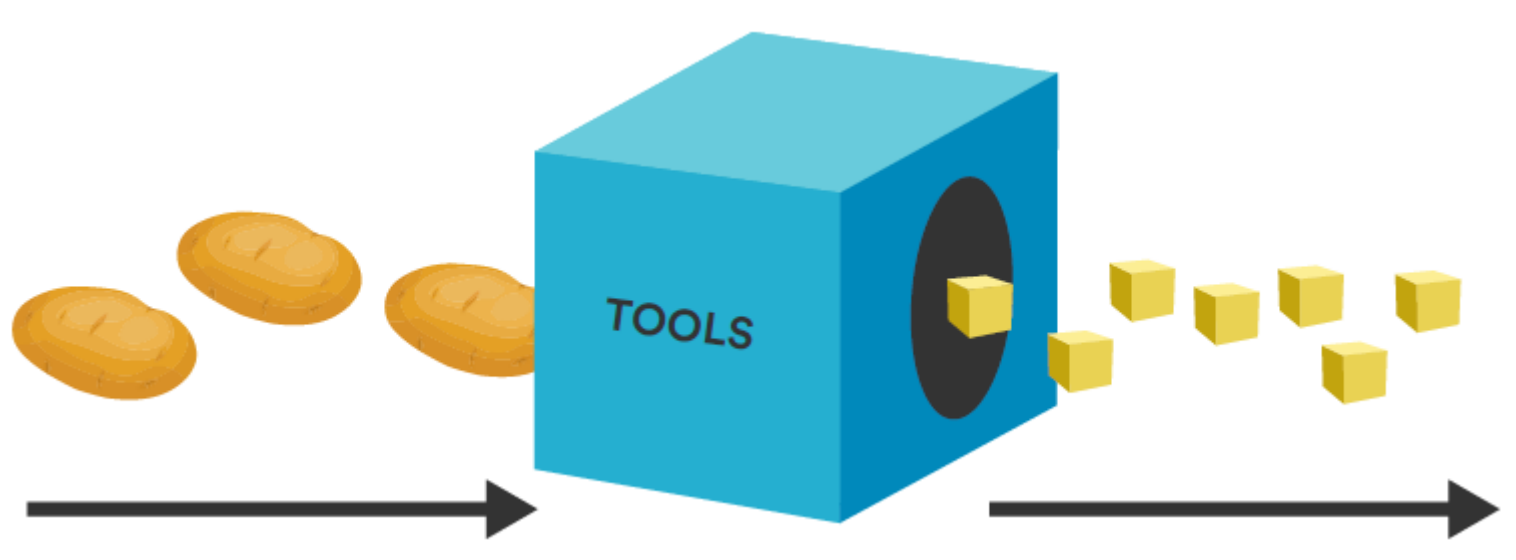
Python Function

Function Introduction

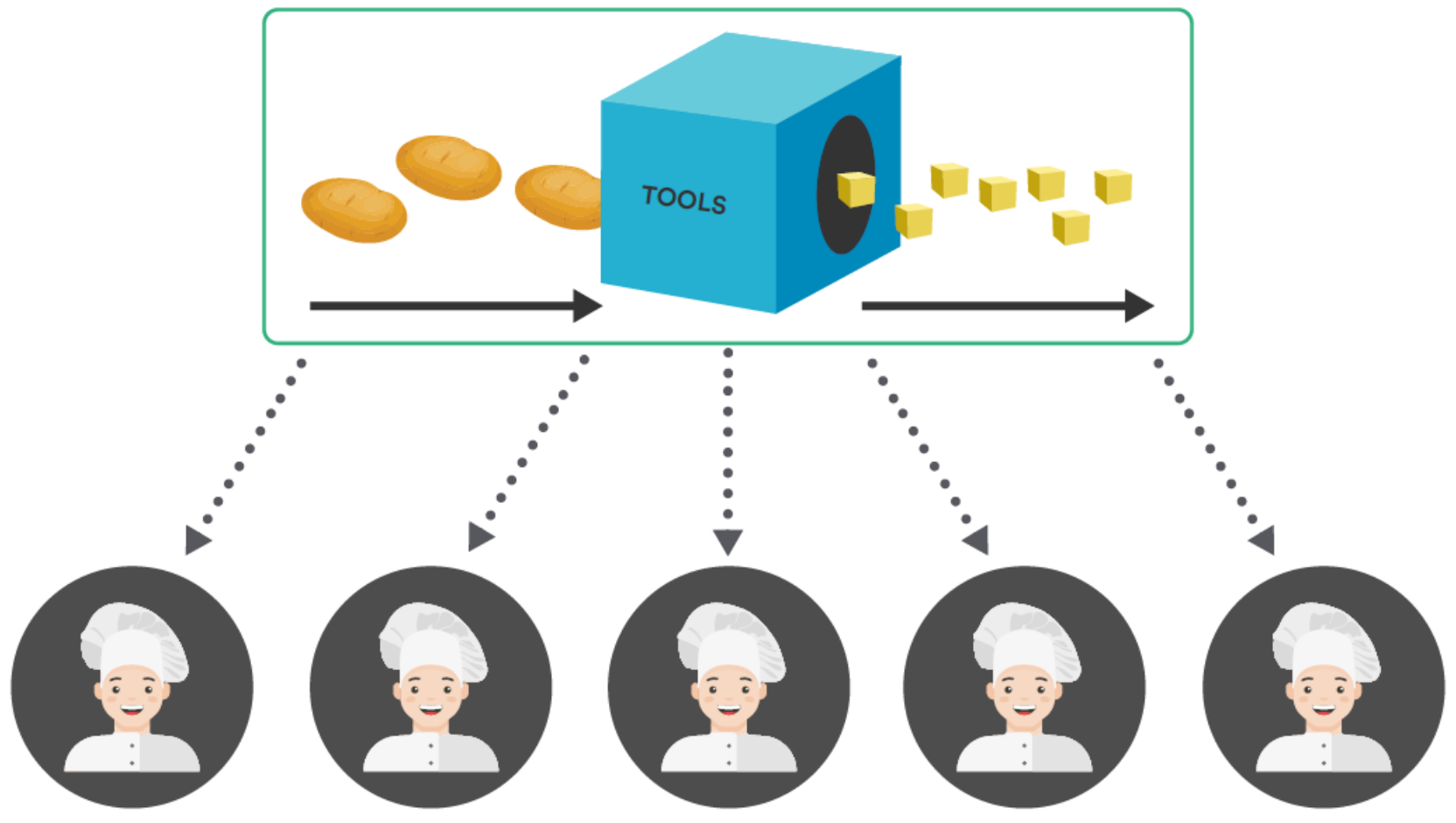
Function Introduction



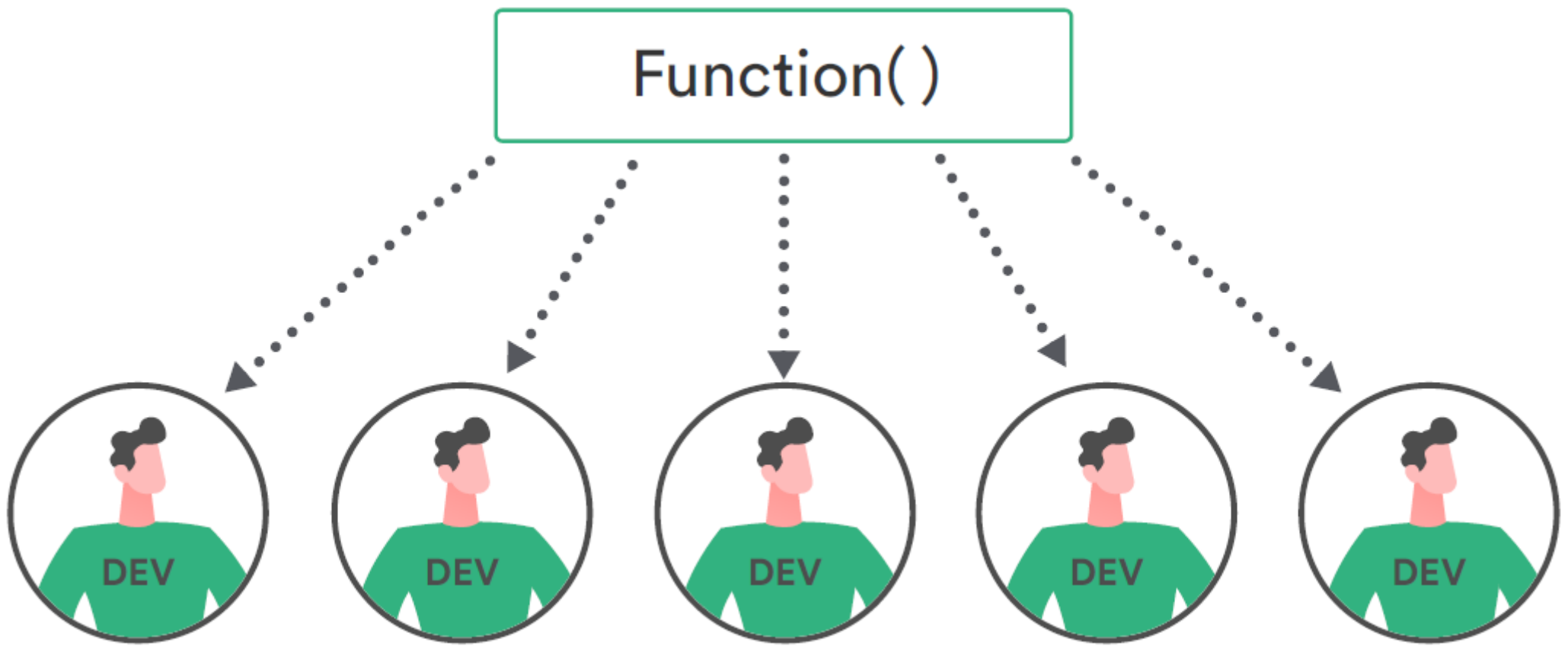
Function Introduction



Function Introduction

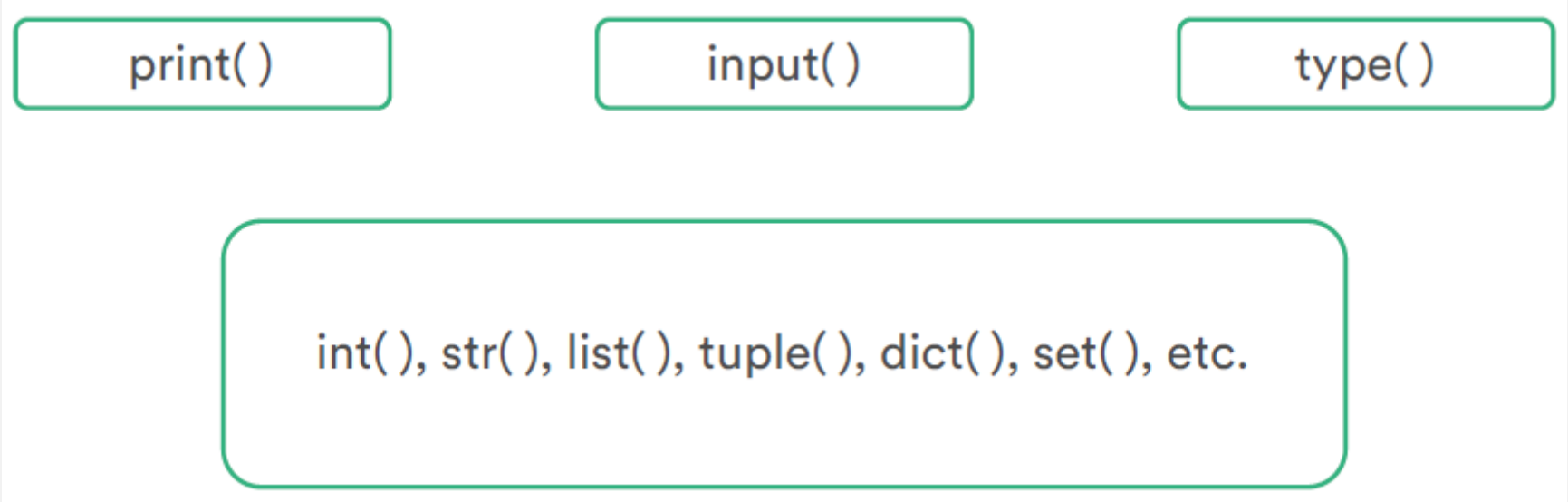


Function Introduction



Function Introduction

Function Introduction



Function Introduction

- What is Function?
- Why?
- When?

Syntax Function

Syntax Function

Regular Function

Function Name Parameters

```
def tambah(angka1,angka2) :
    return angka1 + angka2
```

Keyword def to build Python Regular Function

Keyword return to exit from function, and return value to the caller

The returned value

Lambda Function

Parameters

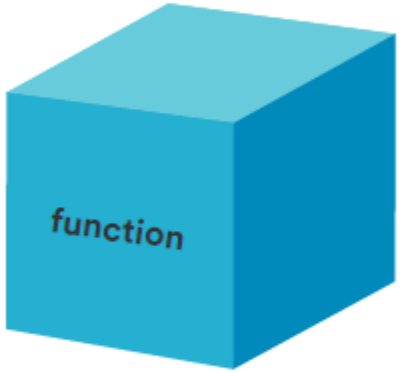
```
lambda angka1,angka2 : angka1 + angka2
```

Keyword lambda to build Python Lambda Function

The returned value

Function Without Input and Output

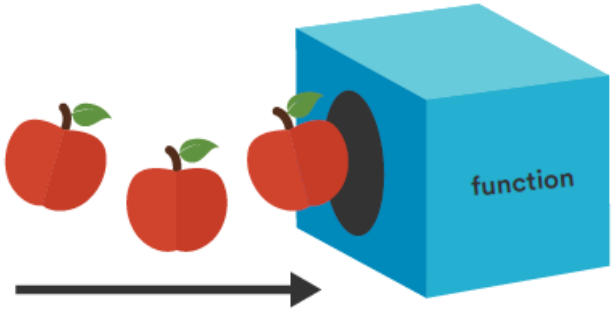
Function Without Input and Output




```
def salam() :
    print('Hello selamat datang di Purwadhika!')
    print('Semoga hari anda menyenangkan!')

salam()
# Hello selamat datang di Purwadhika!
# Semoga hari anda menyenangkan!
salam()
# Hello selamat datang di Purwadhika!
# Semoga hari anda menyenangkan!
```

Function With Input but Without Output

Function With Input but Without Output




```
def salamBalik(nama,usia) :
    print('Hello perkenalkan nama saya {}, dan usia saya {}'.format(nama,usia))
    print('Senang bertemu dengan anda!')

salamBalik('Andi',25)
# Hello perkenalkan nama saya Andi, dan usia saya 25
# Senang bertemu dengan anda!

salamBalik('Budi',27)
# Hello perkenalkan nama saya Budi, dan usia saya 27
# Senang bertemu dengan anda!
```

Parameter or Argument?

```
# Parameter = nama,usia
def salamBalik(nama,usia) :
    print('Hello perkenalkan nama saya {}, dan usia saya {}'.format(nama,usia))
    print('Senang bertemu dengan anda!')

# Argument = 'Andi,25
salamBalik('Andi',25)
```

Parameter Arguments

```
def salamBalik(nama,usia) :
    print('Hello perkenalkan nama saya {}, dan usia saya {}'.format(nama,usia))
    print('Senang bertemu dengan anda!')

# function(parameter=argument,parameter=argument)
salamBalik(usia=25,nama='Andi')

# Hello perkenalkan nama saya Andi, dan usia saya 25
# Senang bertemu dengan anda!
```

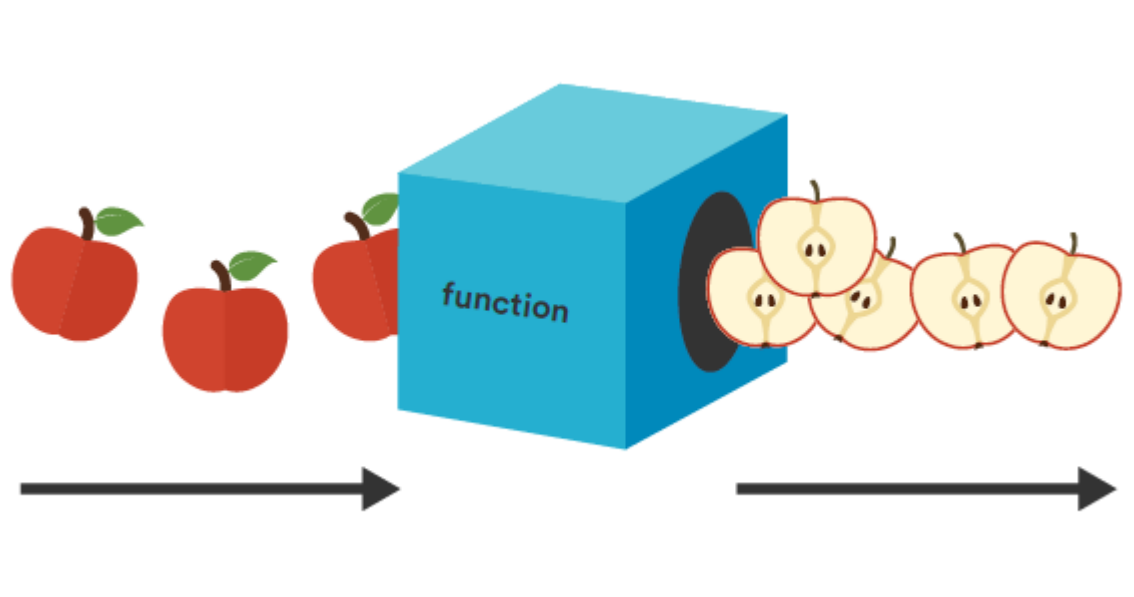
Default Parameter

Default Parameter

```
def salamBalik(nama = 'Unknown' ,usia = 0) :
    if(nama == 'Unknown' and usia > 0) :
        print('Saya berusia {}'.format(usia))
    elif(nama != 'Unknown' and usia == 0) :
        print('Hello nama saya {}'.format(nama))
    elif(nama != 'Unknown' and usia > 0) :
        print('Hello perkenalkan nama saya {}, dan usia saya {}'.format(nama,usia))

    print('Senang bertemu dengan anda!')

salamBalik(usia=25)
# Saya berusia 25
# Senang bertemu dengan anda!

salamBalik('Budi')
# Hello nama saya Budi
# Senang bertemu dengan anda!

salamBalik(usia=21,nama='Andi')
# Hello perkenalkan nama saya Andi, dan usia saya 21
# Senang bertemu dengan anda!
```

Function With Input and Output

Function With Input and Output




```
def tambah(angka1,angka2) :
    return angka1 + angka2

hasil1 = tambah(5,4)
print(hasil1)  #  9

print(tambah(10,2))  # 12
```

Local Variable vs Global Variable

Local Variable vs Global Variable

```
def total(x,y) :
    z = x + y
    return z

print(total(4,5))
print(z) # error
```

- z is local variable in total function, z cant be called outside the function

LOCAL VARIABLE

is defined as a type of variable declared within a programming block. It can only be used inside the code block in which it is declared. Local Variables will exist as long as the code block / function is still under execution. After that, it will be destroyed/deleted automatically.

GLOBAL VARIABLE

is a variable defined outside function. It has a global scope means it holds its value throughout the lifetime of the program. Hence, it can be accessed throughout the program by any function defined within the program, unless it is shadowed.

Local Variable vs Global Variable

1

```
def coba() :
    print(x + 2)
    return x + 3

x = 5
print(coba(), x)
# 7
# 8 5
```

2

```
def coba() :
    x = 8
    print(x + 2)
    return x + 3

x = 5
print(coba(), x)
# 10
# 11 5
```

Local Variable vs Global Variable

1

```
def coba() :
    x += 8
    print(x + 2)
    return x + 3

x = 5
print(coba(), x)
# error
```

2

```
def coba() :
    x = 2
    x += 8
    print(x + 2)
    return x + 3

x = 5
print(coba(), x)
# 12
# 13  5
```

3

```
def coba() :
    global x
    x += 8
    print(x + 2)
    return x + 3

x = 5
print(coba(), x)
# 15
# 16  13
```

Callback Function

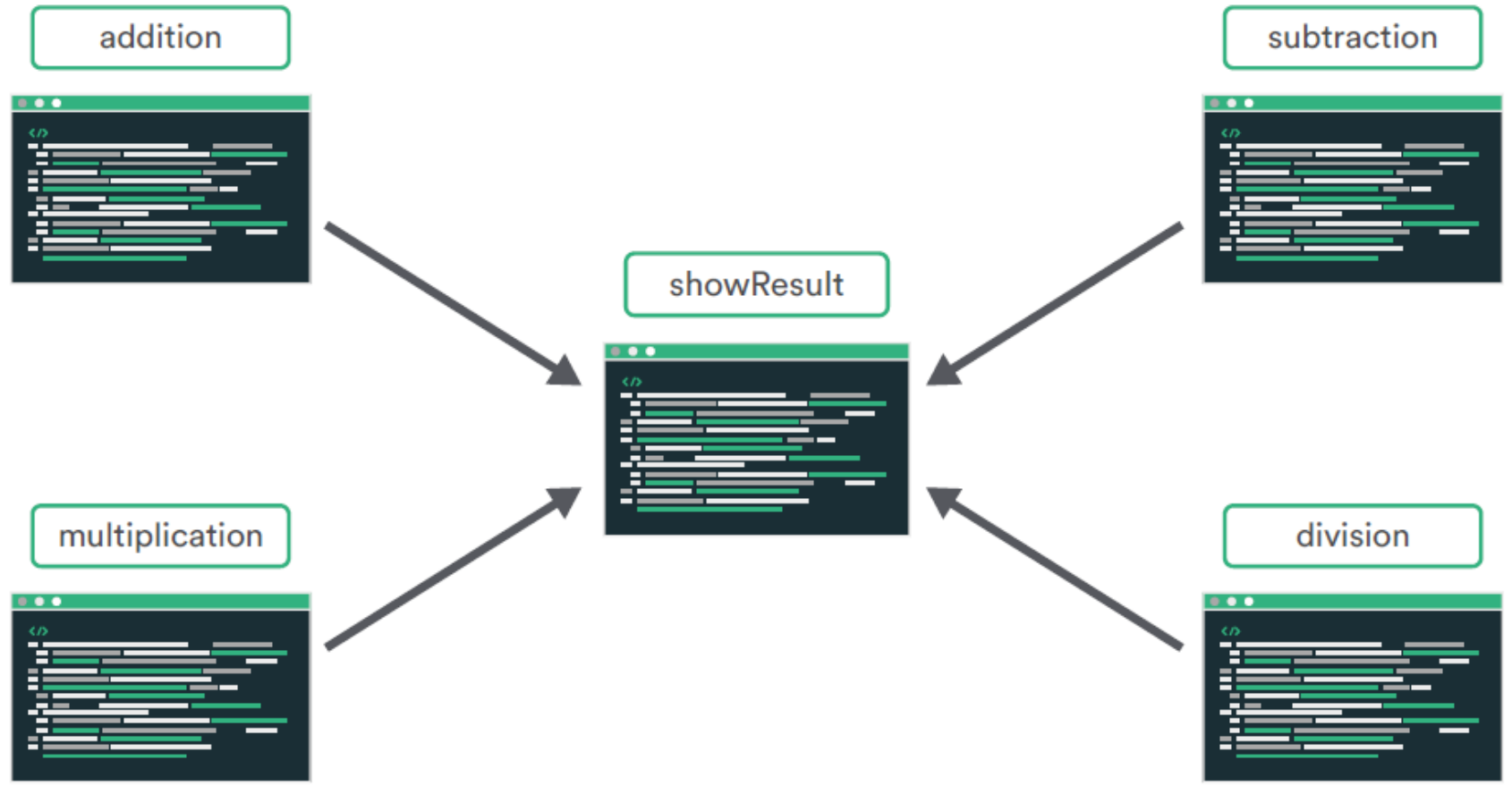
Callback Function

Callback Function

```
def tambah(num1,num2) :
    return num1 + num2

def kurang(num1,num2) :
    return num1 - num2

def hitung(fnOperator, angka1,angka2) :
    hasil = fnOperator(angka1,angka2)
    return hasil

print(hitung(tambah, 9, 5)) # 14
print(hitung(kurang, 9, 5)) # 4
```

Calling Other Function

Calling Other Function



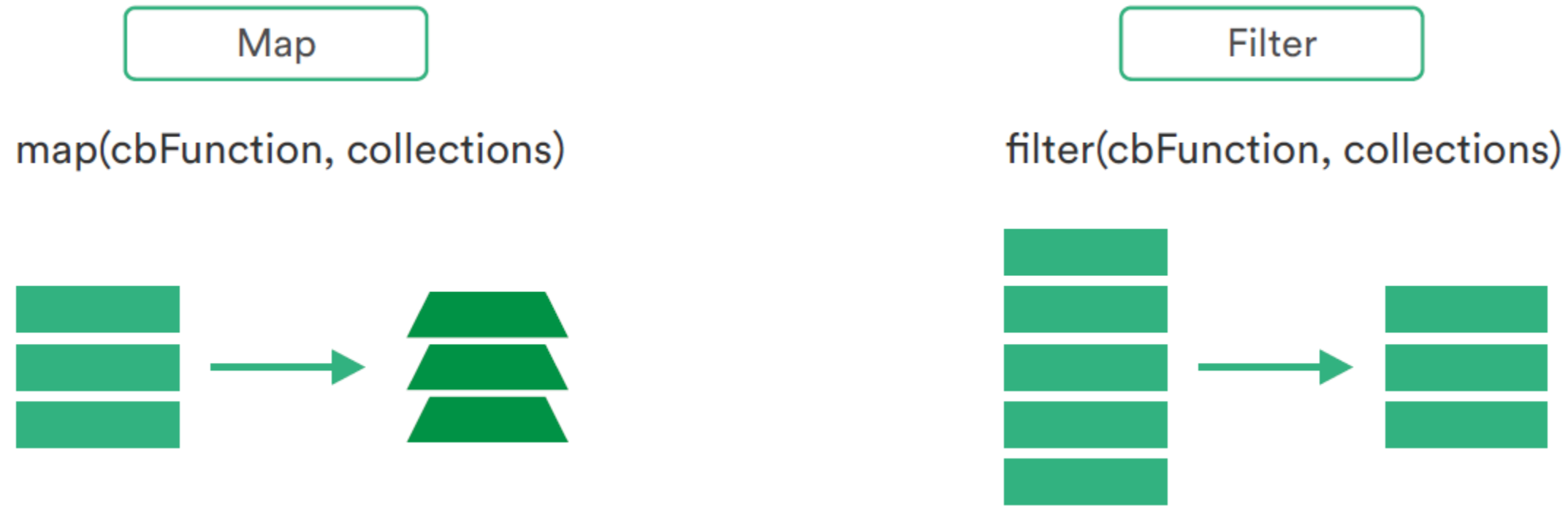
Recursive Function

Recursive Function

```
def countdown(counter) :
    print(counter)
    counter -= 1

    if(counter >= 0) :
        countdown(counter)

countdown(3)
# 3
# 2
# 1
# 0
```

Python Map and Filter Function Introduction

Python Map and Filter Function



Python Map Function

Python Map Function

"Changing form of value"

[1, 2, 3, 4, 5] → [2, 4, 6, 8, 10]

[1, 2, 3, 4, 5] → [0.5, 1, 1.5, 2, 2.5]

[1, 3, 4, 5] → ["ganjil","ganjil", "genap", "ganjil"]

Lambda Function

- A lambda function is a small anonymous function.
- A lambda function can have any number of parameters, but can only have one expression.

Syntax

lambda parameters : expression

Lambda Examples

```
kali2 = lambda angka : angka * 2
print(kali2(7)) # 14

tambah = lambda angka1, angka2 : angka1 + angka2
print(tambah(15,2)) # 17

kurang = lambda angka1, angka2, angka3 : angka1 - angka2 - angka3
print(kurang(20, 4, 6)) # 10
```

Python
Map Function

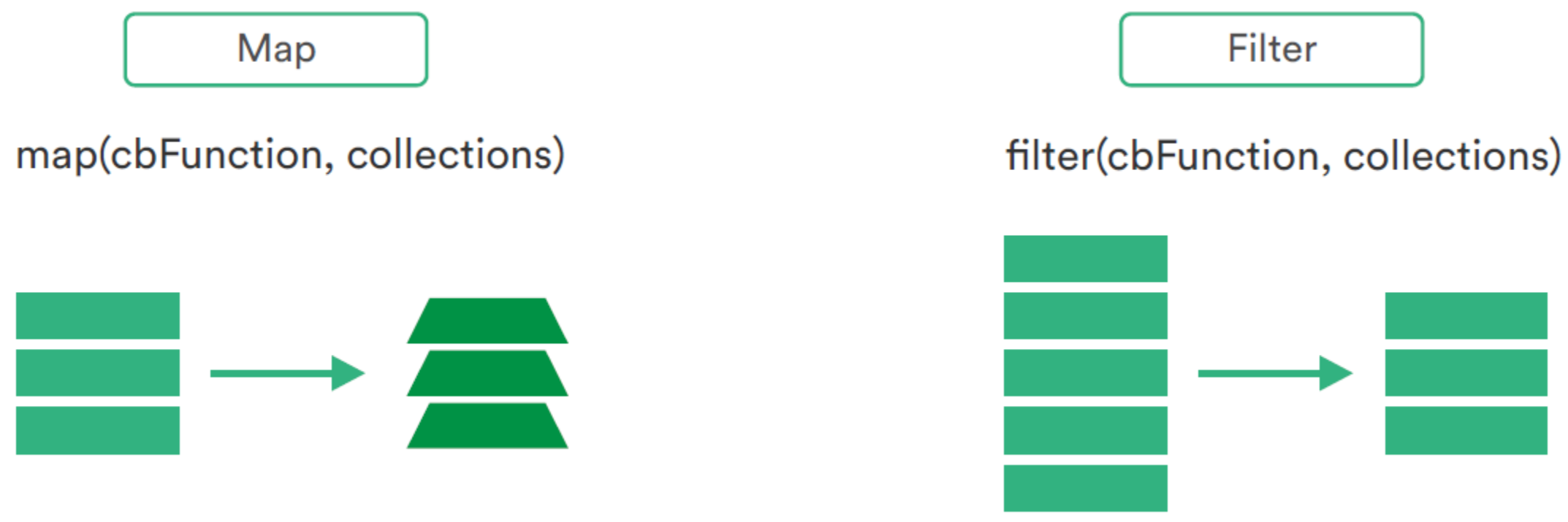
```
list1 = [1,2,3,4,5]

def kali2(angka) :
    return angka * 2

hasilMap = map(kali2, list1)
print(hasilMap, type(hasilMap))
# <map object at 0x0000027509756640> <class 'map'>

hasilMapList = list(hasilMap)
print(hasilMapList)
# [2, 4, 6, 8, 10]

hasilMapList1 = list(map(lambda angka : angka * 2, list1))
print(hasilMapList1)
# [2, 4, 6, 8, 10]
```

Python
Map Function

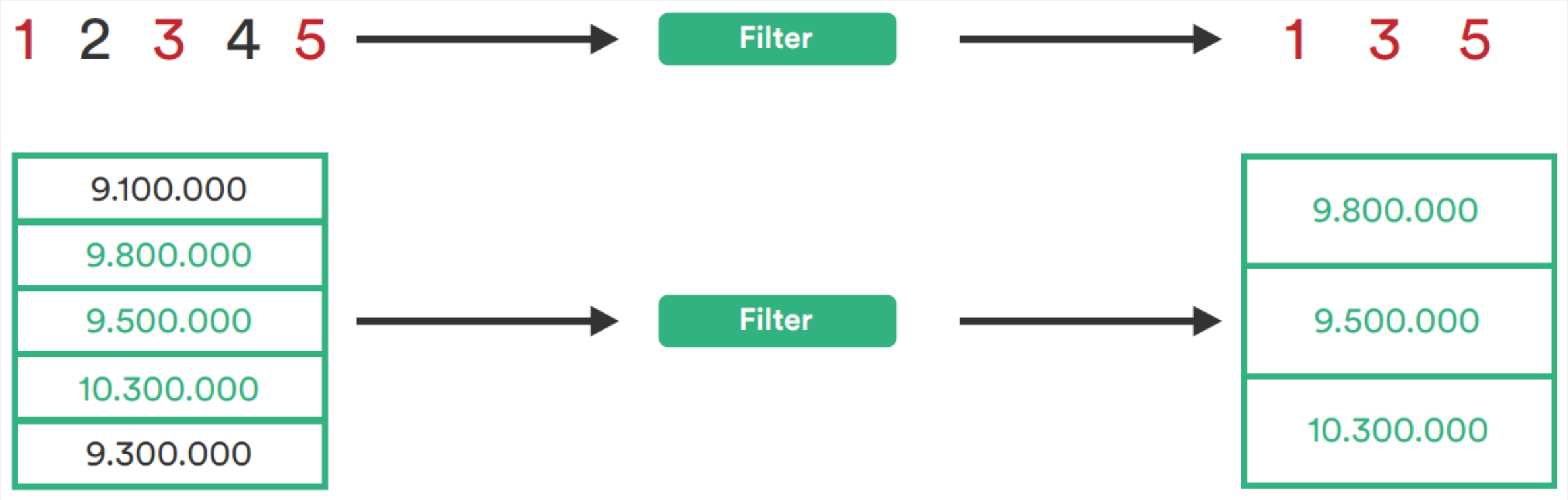
```
def mapDuplikat(fn,collection) :
    newCollection = [ ]
    for item in collection :
        newCollection.append(fn(item))

    return newCollection

list1 = [1,2,3,4,5]

def ubah(angka) :
    return 'Angka { }'.format (angka)

newList = mapDuplikat(ubah,list1)
print(newList)
# ['Angka 1', 'Angka 2', 'Angka 3', 'Angka 4', 'Angka 5']

print(mapDuplikat(lambda angka : 'Angka { }'.format(angka), list1))
# ['Angka 1', 'Angka 2', 'Angka 3', 'Angka 4', 'Angka 5']
```

Python Filter Function

Python Filter Function



Python Filter Function



Python Filter Function

```
list1 = [1,2,3,4,5]

def angkaGenap(angka) :
    return angka % 2 == 0

hasilFilter = filter(angkaGenap, list1)
print(hasilFilter, type(hasilFilter))
# <filter object at 0x000001199E6E6640> <class 'filter'>

hasilFilterList = list(hasilFilter)
print(hasilFilterList)
# [2, 4]

hasilFilterList1 = list(filter(lambda angka : angka % 2 == 0, list1))
print(hasilFilterList1)
# [2, 4]
```

Python
Filter Function

```
def filterDuplikat(fn, collection) :
    newCollection = [ ]
    for item in collection :
        if(fn(item)) :
            newCollection.append(item)

    return newCollection

list1 = [1,2,3,4,5]

def angkaGenap(angka) :
    return angka % 2 == 0

newList = filterDuplikat(angkaGenap, list1)
print(newList)
# [2, 4]

print(filterDuplikat(lambda angka : angka % 2 == 0, list1))
# [2, 4]
```

Exercise

Google Search Console
Couverture > Introuvable (404)
Vue d'ensemble
Inspection de l'URL
Performances
Résultats de recherche
Discover
Index
Couverture
Sitemaps
Suppressions
Améliorations
Signaux Web essentiels
Ergonomie mobile
Logos
Extraits d'avis
Toutes les pages connues
Introuvable (404)
État : Exclues
EN SAVOIR PLUS
Pages concernées
52
Exemples
URL
MacBook Air

Soal 1

Melanjutkan Aplikasi Market

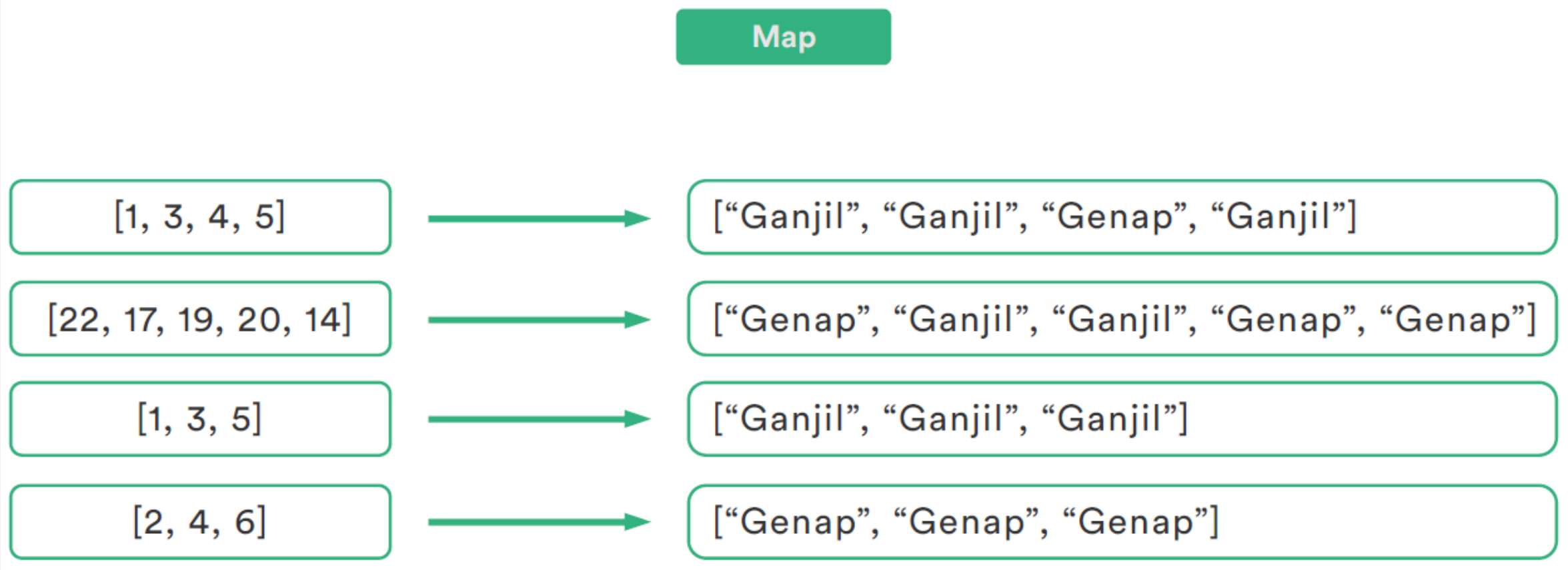
Soal 2

Masukkan 5 Film Kesukaan anda dipisahkan dengan koma : Avengers,Hulk,Iron Man 3,Wonder Woman,Batman

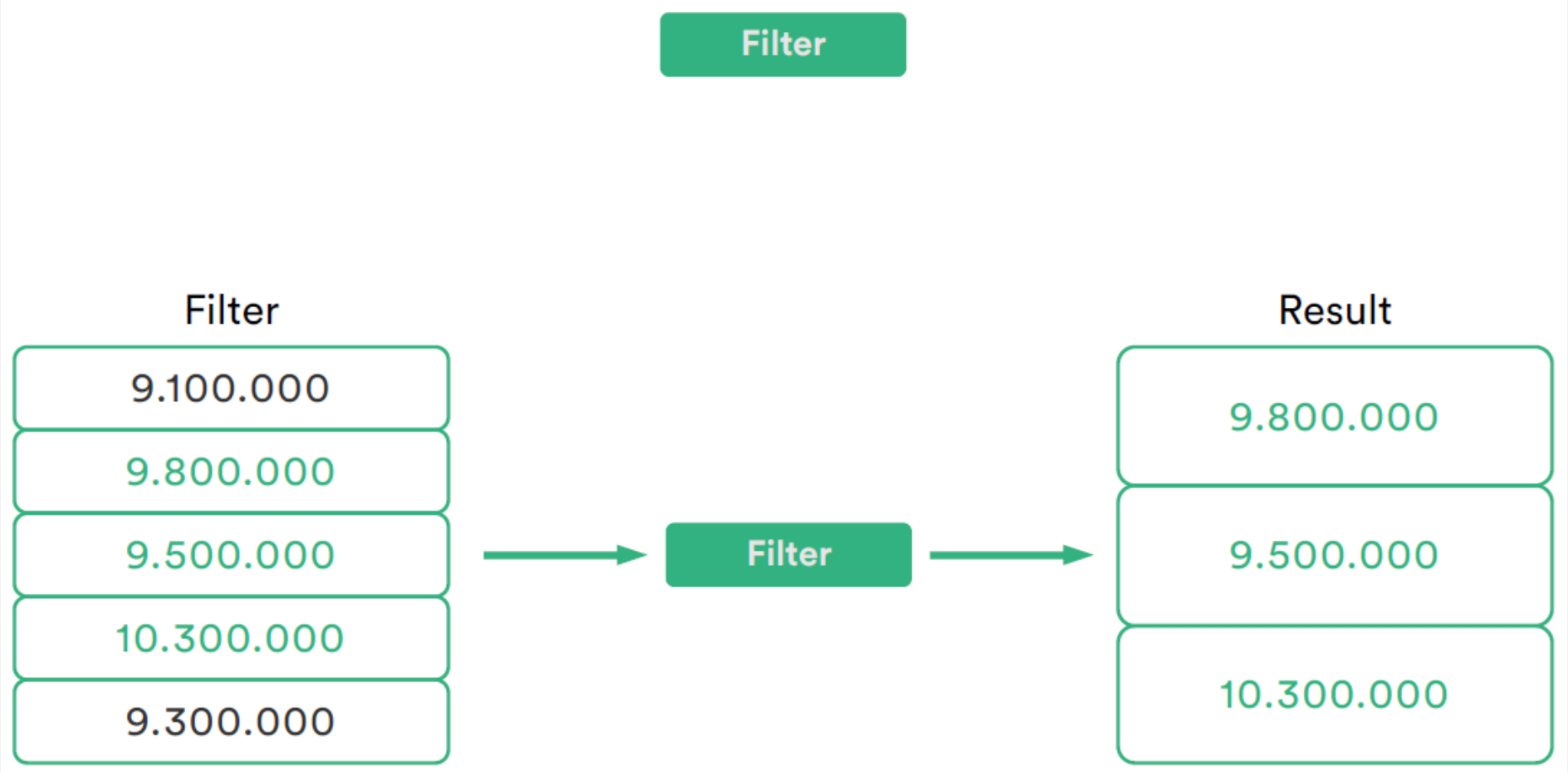
Masukkan 5 Film Kesukaan teman anda dipisahkan dengan koma : Avengers,Hulk,Iron Man 2,Superman,Wonder Woman

Kesukaan Film kalian yang sama sebesar 60.0%

Soal 3


Map
[1, 3, 4, 5]
["Ganjil", "Ganjil", "Genap", "Ganjil"]
[22, 17, 19, 20, 14]
["Genap", "Ganjil", "Ganjil", "Genap", "Genap"]
[1, 3, 5]
["Ganjil", "Ganjil", "Ganjil"]
[2, 4, 6]
["Genap", "Genap", "Genap"]

Soal 4


Filter
Filter
9.100.000
9.800.000
9.500.000
10.300.000
9.300.000
Filter
Result
9.800.000
9.500.000
10.300.000

Soal 5

Lanjutkanlah soal market yang sudah kita kerjakan sejauh ini. Pecahlah code-code yang terdapat pada program market kita ke function masing-masing supaya lebih rapi dan lebih mudah untuk dimaintain.

Thank You!

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